How do rings form? One possibility is tides: If an errant world passes close to a planet, the interloper's near side is gravitationally pulled toward the planet more than its far side; if it comes close enough, if its internal cohesion is low enough, it can be literally torn to pieces. Occasionally we see this happening to comets as they pass too close to Jupiter, or the Sun. Another possibility, emerging from the *Voyager* reconnaissance of the outer Solar System, is this: Rings are made when worlds collide and moons are smashed to smithereens. Both mechanisms may have played a role.

The space between the planets is traversed by an odd collection of rogue worldlets, each in orbit about the Sun. A few are as big as a county or even a state; many more have surface areas like those of a village or a town. More little ones are found than big ones, and they range in size down to particles of dust. Some of them travel on long, stretched-out elliptical paths, which make them periodically cross the orbit of one or more planets.

Occasionally, unluckily, there's a world in the way. The collision can shatter and pulverize both the interloper and the moon that's hit (or at least the region around ground zero). The resulting debris—ejected from the moon but not so fast-moving as to escape from the planet's gravity—may form, for a time, a new ring. It's made of whatever the colliding bodies were made of, but usually more of the target moon than the rogue impactor. If the colliding worlds are icy, the net result will be rings of ice particles; if they're made of organic molecules, the result will be rings of organic particles (which will slowly be processed by radiation into carbon). All the mass in the rings of Saturn is no more than would result from the complete impact pulverization of a single icy moon. The disintegration of small moons can likewise account for the ring systems of the three other giant planets.

Unless it's very close to its planet, a shattered moon gradually reaccumulates (or at least a fair fraction of it does). The pieces, big and small, still in approximately the same orbit as the moon was before the impact, fall together helter-skelter. What used to be a piece of the core is now at the surface, and vice versa. The resulting hodgepodge surfaces might seem very odd. Miranda, one of the